Advanced Matplotlib Transcript

We'll now continue our understanding of advanced features of Matplotlib. In this lesson, we will try to understand the different native plots that are in Matplotlib and how we can use them for our data visualisation tasks. In addition to the NumPy and the Matplotlib, now we'll be also importing the Pandas library for our work.

So let us just copy the code, go to the work environment and start running. Pandas is also now imported into our workspace. So now let us try to understand how to use a histogram.

Histogram is used for visualising the distribution of a single numerical variable and what is the frequency distribution, what are the different categories and how many frequency counts are there in the particular class interval. So the function to plot histogram is plt.hist and you have to give the data and number of bins, so on and so forth. Now, let us take quickly the example.

So I'm just going to copy the code. I'll explain each line in the code. So np.random seed is a random number generator where you are trying to fix the particular random sequence.

And then you are using that as the random data. This is a random data of thousand variables, thousand random numbers are generated from a standard normal distribution. And we are fixing the configuration of random seed so that every time you run you get the same data for the graph.

So then you are using the plt.hist. So once you hover on the particular command, you get the entire description in the box, which you can examine closely with the examples and try to use those features and build more effective visualisation. See, we might not be able to explain each and every feature or the syntax elements in a particular command, but the students are asked to explore these commands and get back to us if there are any doubts or hiccups in the way in exploring such features. OK, now we are just using the plt.hist, which is the function for drawing the histogram or asking the system to plot a histogram.

We are giving data which is stored in this data variable. And we want 30 bins, 30 class intervals. Colour is sky blue and then edge colour is black.

Edge colour is the edge colour of each histogram box and title is histogram of a normal distribution. And you have to label the x value as value and y is as frequency and finally show the histogram. So histogram of normal distribution, the title, you have the figure, you have the values 0 to 4 and 0 to minus 3. Then you have 30 bins.

These are 30 bins in data. Then the x is value and frequency. So 0 occurs.

Right. So 0 occurs almost 100 times. Now, let us see how to add labels to this.

OK, so I'll just ask Gemini here how to add labels in histogram. So that we will know the number of times 0 is occurring, number of times 1 is occurring or any value. What is the length of each bar instead of going to the axis? So, OK, so it is saying, yeah, so add labels hist.bins and histogram bins is equal to 30.

OK, so it is giving this particular code where you can use the label. Oh, sorry. Yeah, so this is how you are getting the labels.

OK, so your 0 value is occurring almost 24 times, something greater than 0, but between 1 is 1 or 2, you can get the labels. And here only labels with count greater than 0. This is all it is using. It is using a for loop and then getting the labels.

OK, again, you can go here, go to the examples and take, for example, here. So statistics, then you have a histogram here. So you have the description of a histogram.

OK, so you have the various features of histogram here. So you can try to show the histogram plots and then try to understand the different features of how to plot a histogram. OK, we have something called by histogram also.

So it shows the distribution on both sides. So please explore all these features at your convenience. So now next we move to the bar plot.

So the purpose of a bar plot is to compare two different discrete categories. So in earlier example, we have seen how to compare the different types of fruits and their production numbers. Now we are trying to understand what is the different social media apps and what are the users in millions that visit these apps on a monthly basis.

So again, the function for plotting a bar plot is plt.bar. You have labels and values. What are labels? Labels are the labels for different categories. For example, in our case, the platforms or social app connect me, share it and link up and the names will become labels and the users is the data.

So in bar graph, you should have categorical variable and the quantitative variable, which shows the number of counts that are there on each of these categories. So what are we doing? First, we are creating data. So simply we are just copying the variables.

So first variable is platform or the array, which connects the four types of which has the four types of social media apps. And then the next is user in millions, which shows the corresponding number of users in millions for each of these apps. So I've defined these two variables.

Then I'm creating a data frame using these two variables. So data frame to create a data frame, you use df is equal to df or you can name it anything. pd.dataFrame is the function in which you can define the data frame.

What does it create? It creates a matrix of columns and rows. So what are the columns here? You have columns of platform and you are defining the platform as the column name. And what is the data in that? It is coming from this array platforms.

Then you have users. It is and it is taking data from this array of users in millions. And then this data frame is getting generated.

Now you can also use df.show. So just printing the df, you will see how the data frame is created. So in Python, you can just use this suggested chat function, which is also show the different types of plots that are there. But our purpose here is to understand how to use this data for plotting one particular type of graph that is bar chart.

So we are using ex is equal to df.plot. So we are saying that data frame has to be plotted. And what is the kind of plot we want? Bar. Then we want x variable to be platform, y variable to be the number of users.

You want legend to be false because there is no need for legend. Colour to be cornflower blue. OK, so simply this is what you have got.

OK, so let us do this again. Just trying to and I am also using the labels and setting the labels here. What is the label title? I am saying monthly active users by platform, x axis is platform, y axis is users in million.

And set x tick labels. OK, and I want to have the labels to be rotated horizontally instead of vertically. This is 90 degrees.

I want to know it to be 0 degrees and also make it I think. Let us do the 45 here and keep labels. Let us see how does it work.

See now it is at 45 degrees. It is otherwise if you make it zero, it would be horizontal. OK, yeah, let us look at the df.plot. So what are the parameters that you can give here df.plot? So vertical, we have line plot, bar for vertical bar, bar h for horizontal, hist for histogram, box plot, kernel density estimation.

Scatter plot, pi for ply area. You can use all these features to call different plots. OK, this you might have to remember which is the function to call for different type of plots.

So instead of vertical, if you want to just do horizontal, you have to do bar h. Right, so then you have the sequence of numbers here. OK, what are the other points you can try to do? OK, figure size, layout, legend, grid. So all this you can try to use colour map.

OK, position. So all this you can use for creating more features. You can embed more features into the graph.

OK, so you can use a secondary y-axis, secondary x-axis. I mean, you can try to plot two series. OK, you can do all that like in case of in Excel, but practise this kind of things by exploring more data examples.

OK, yeah. So I'll just do one more thing before we move to the other. I am trying to do it as bar h, bar horizontal.

Same graph now you will have like this. OK, yeah. Now let us see the next bar plot.

It is grouped bar plot with error bars. The purpose here is to compare multiple series across categories, often including error bars to show the measurement uncertainty. And this method is easily achieved using Pandas.

df.plot is equal to kind again is bar. But here we will have y error to be true. So that's how we'll use the feature to plot the grouped bar plot.

Let us take the example and execute it step by step. In this case, again, I'm creating a data frame before and after. OK, engagement scores after website redesign.

Before it was 6.5 and 5.8. After it is 8.2 and 6.1. Index is redesign and control group. So what has happened to the group that has shown the redesign side? The way to read the data is the users were divided into two groups, the experiment group and the control group. So the experiment group or the treatment group was shown the redesigned website and their engagement has increased from 6.5 to 8.2. Whereas the control group has slightly increased from 5.8 to 6.1. Therefore, one can conclude that the redesign of the website has increased the engagement of the participants.

OK, and there is a measurement error. So the measurement error is 0.05 and 0.04 in the earlier instance that is before and after it is 0.03 and 0.5. So now what are you trying to do? You are trying to plot the before and after as graph as bar graphs. So this is for control group.

It will be 6.5 and 8.2, 5.8 and 6.1. So this will be stacked against adjacently with the error bar on the top. So you have again in the last like in last case you are using df.plot. You are plotting the data frame, right? And then you are plotting the data frame. And then the type of bar you want is vertical bar.

So kind is equal to bar. But here y error, the y error is y-axis variable error is true. You want to show the error.

Caption size is 4. Colour is Salomon and light blue for different categories. Title is user engagement before and after redesign. Average engagement score.

And again you want x-axis labels. And where are you getting this from? You are getting from df.index, which is redesign and control group. These are the two categories.

Then you have rotation. You do not want them to be vertical. You want them to be horizontal.

And finally you are showing the plot. So this is how it will look like. You also have the legend which shows before and after.

So before redesign and after redesign. Before redesign and after redesign for control and redesign groups. And average engagement score with the error bars.

So the error bars also show the confidence with which you should take the average engagement score. So it is clearly showing even if you account for the error bar, there is a serious increase in the, I mean significant increase in the engagement activity after redesigning the website. So this is how you can do a grouped bar chart.

Again I am saying the examples are, there are so many examples. So we cannot discuss each one, every one of them. So please, so this is the error bar example.

So you can also explore this in detail from the Mathplotlib source documentation. And try to experiment them and get back to us in case if you face any problems. Now we will next move to pie chart.

Pie chart is a very interesting and important chart to show the composition of different categories. But it is useful only if you have limited number of categories so that you can show the, you can give a very visual appeal to your audience if there are less categories. But pie chart should not be used if there are too many categories.

It will become really difficult to perform the detect and compare functions. Again you can use plot.pie or if you are using in a df, you can use df is equal to, kind is equal to pie. You can do both ways.

So now let us see the example. So in this case we are taking the sources, website traffic sources for a month. Organic search, direct traffic, social media and there are five sources that you have categorised.

And what is the percentage organic search is 45, direct is 25, 15 is referral, 10 is social media, 5 is other. Okay. So this is what you have gathered the data.

So these two source and the traffic percentage, you want to plot that as a pie chart. So you have determined the figure size. For a minute, please ignore or I'm just going to annotate this particular explode portion.

So let us talk about that later. So first we are trying to understand the data. You are five different sources of your website traffic and you have estimated the percentage of users that are coming through each channel.

You want to present it as a pie chart. So plot.figure is equal to figure size 6,6. Right.

So it is always better to have a pie chart in a square shape because it fits and gives an aesthetic appeal for your pie chart. So we are using plot.pie function. Okay.

So it says what are the different features that you can use in plot.pie. Okay. So what is the main element that pie is plotting? So it's a circle divided into different sectors. These sectors represent the proportion for a particular category.

So first is your traffic percent. That is the main variable. And you want to label them.

What is the each sector representing? So labels are sources in one to one correspondence has to be ensured. The order of your variables and the order of the numerical values are different. You might end up having wrong results.

So be careful. The order in which your sources are labelled are there. The same way the numerical values should be there.

Or if your numerical variables are ordered ensure your labels are ordered in that particular fashion. Then we will talk about explode. We have not defined explode here.

For a moment I am going to take this off. Okay. I just want to show this later.

Okay. Then start angle is at 90. Rotate at the start of pie shadow is true.

You want to show this shadow at the back. So title. So yeah.

We're just going to run this. So this is how you get. So let me explain.

So this feature at the end is called the shadow. So I'm just removing shadow also. Just trying to run this again.

So now this is more clean. Okay. So remember please avoid 3D effects if it is not required.

Now let us explore the explode feature. What I want to do is I want to get one of these parts little away. So that it creates an impact showing this is the particular element that I want to focus on.

Maybe you are doing let us say campaign on social media to increase the visibility of your website. So you want to highlight only the social media as a pie that is drawn out as a cake piece that is pulled out of this pie. So let us do that.

For that this explode function will be helpful. So what is explode function is doing. So if you see at the export function it has five elements representing each one of the sector or the particular label and you can do zero values.

Okay. And here I'm including the explode is equal to explode. You have to give the explode is an option which explodes the graphic and you have to give the share at which the distance to which it will explode.

So I'm just giving all 0 0 0. And our social the website is social media is the fourth one. So 1 2 3 4 I am just putting it here 0.02. So and I'm just running the thing. So you can see that social media is now moved as a bit as compared to others to showcase and get your audience attendance on that part.

Okay. So this is how you can use the explode feature along with other features you can colour that make it more evident. If you want to highlight one particular sector and then from there build a story for your data visualisation exercise while using the composition for a pie chart.

Right. So this is one more advanced feature. Now let us go and look at the radar chart.

So again radar chart has a very distinctive usage. It helps to display multivariate data that is multivariable data on an axis starting from the same point. So it is always used to compare multiple quantitative variables.

So for example one of the most used cases a radar chart is a financial stability map. Let me quickly show the example which Reserve Bank of India uses. Okay.

So banking stability map. So this is a radar plot where you have five different qualities of banking and you try to all of them start at the same origin and extend in different dimensions. By plotting them together on the same plane you will be able to showcase how as a phenomenon together all these variables are affecting in this case banking stability.

So likewise if you have a multivariate variables which start at the same origin and extend in the same direction but together they represent a phenomenon you can use a radar chart. Okay. So a simple example would be you are a store doing retail sales.

There are seven or eight product segments and you want to understand in each segment how much target have you achieved. Conventional way would be to present the target achievement in a tabular form. But if you present as a radar chart the area covered under the radar can be easily understood as the percentage achievement across all dimensions.

And if you have time series like two three periods you can see whether it is increasing coverage or decreasing achievement of targets. Okay. So this is one of the use cases.

So let us quickly see how to do the radar chart. Again take the port go to the port snippet right paste it here. So theta is the polar variable or the radial variable that I am generating using this function 0 to 2 into np into pi.

np is where panda sorry numpy pi is the variable 3.14 and you are having this angle. It is a radial variable. It generates number from 0 to 360.

So then you have. So you have trying to do a rose curve equation. So let us not worry about the math part here.

We are trying to showcase the main feature is if you have the seven dimensions you can think of seven product lines of your retail store or eight product lines. You are trying to understand how each one of them is performing in a single snapshot. Okay.

So using these two I am just running the data and plot size again should be a square shape because it is also like a pie. Then you have subplot projection is polar because using radial coordinates. Right.

Then you have the theta which is the and the radius colour we are using purple line with these three. Okay. And we are running the example.

So this is what we are getting. So here the main point is theta generates the angle. Okay.

Radius is the length of this particular. Where is it located. Six.

So now if I make it into if you have six categories if I have eight categories. Okay. So then I get eight lines.

Okay. This is how it is function. Okay.

Yeah. So. And let me do sign just for fun.

Same. And. Okay.

So this is interesting. So once we do change the function cosine or sine we will get different object shapes. Okay.

Yeah. Now we will move to the most important part of any visualisation exercise is to inducing text elements into your graphs. So if you remember one of the most common used and most powerful types of attentive attributes are visualisation types is using of simple text.

So when you have a simple text you can you want to bring the attention to a particular aspect of your graphic. The best case is to use a text box and try to grab the attention of your audience to that particular point in the graphic. Okay.

So let us try to bring in text elements into our graphics and see how they can be used for improving the attention span of our audience or driving our attention of our audience to a particular point on the text. So again I am using this very simple example. Okay.

We can go to Matplotlib. Yeah. So Matplotlib.

Let's come back here. So text labels and annotations. You have an entire holes range of examples on how to use text in the graphic.

We are taking a very small example and trying to explain. But again I urge you to explore this very deeply so that you know how to use and customise text into your embedded text into your graphs. So what am I doing.

I'm just creating a linear space of X 0 to 10 using the sine X function and plot X comma Y. OK. I'm running the selection. You should get a graph that you have seen in the initial stages of the class.

It's a sine X function plotted from 0 to 10 of X and it varies from 0 to 1. 0 1 minus 1 0 1 minus 1. That's how a sine X function goes. So I'm also adding a title to this. OK.

So plot of sine X. But it has to be done in sequence. So this is what I've done. Now I'm adding plot dot text at 3 comma 0. OK.

3 comma 0. And saying that it is simple text font size is 12. Let us see where it will get added. 3 comma 0 is 3 is X axis 3. That is somewhere here and Y axis 0. That is somewhere here.

You should get the text here. OK. So this is what I'm trying to do.

I'm just running this selection again. So you got the text here. Now I just go back and change this to.

OK. 2 comma minus 0.5. OK. So see the text is coming here.

OK. You can type the text. So instead of simple text I will say data visualisation.

OK. So that's where you get data visualisation. Right.

Now add a row annotation. So we are trying to annotate that with a row. OK.

So again just simple do this with simple text. Right. So peak X is 1.5 times NP dot pi and peak Y is NP dot sin of peak X. So we are defining X as 1.5 times number of pi.

OK. 1.5 times the pi value and Y is sin of peak X. OK. And you want to annotate that as local minimum which would be here and then you want to say that it is a local minimum.

So you are trying to use this plot X comma Y is equal to peak X comma peak Y. Text is this particular thing and you are trying to show it is in black colour and you want it to be in centre. So I am just running this again. So you get a local minimum indicated with an arrow.

So how is this arrow coming? It is coming with arrow props. So it is showing arrow props. You can use this function with arrow props and then show the particular point on the graph.

Yeah. So now let us move to the use of style sheets. OK.

So style sheet is a particular template on which the pre-attentive attributes are already fixed. For example the background colour, the grid, the line width, the colours. All these things are predefined in a style sheet.

It is a template which is already designed for you so that you can use them for colouring getting those attributes onto your graph without worrying about which colour and what pattern to use. So it's a predefined template that you can use. So let us try to understand there is one template called 538 and there is one more template called C1 underscore V0 8. So these are two style templates that you can run and get the predefined styling patterns onto your graphic.

OK. So let us first get this 538 template. So I am just using the plot dot style dot use.

This is a function to use a style sheet and we have to name the style sheet as these. OK. So what are the parameters? Style dot use the moment you say.

OK. You can name it as 538. It is a predefined style sheet.

Then I am just creating a linspace and I am using two functions. Functions are not important. You can simply use cos x and y x also.

OK. And then I am trying to label them as series A series B and using the title plot with 538 style and with the legend. So let me I think the better way to do this would be.

Yeah. So let me do this. First show a simple plot then apply the style sheet.

Yeah. So let me just walk this through. OK.

So I am just plotting a simple function of x and y using cos and sin functions without any style sheets. So this is how it would look. But now I want to use a predefined style sheet.

First I am going to apply the style sheet and then plot the graph. OK. So the same graph I am just plotting with the style sheet.

So first I am invoking the style sheet with this particular function. Then I am running the entire plot function. So I would get a predefined style sheet.

Please note the difference between the simple graph and the stylised graph you have. OK. And to return back to the normal you always have to use this function PLT reset to defaults.

So if you don't do this and keep running the code you would get a stylised graph only. So to reset at original you have to use the PC PLT.OC defaults. Otherwise you can simply use the style sheet and then plot the graph.

You would get a stylised graph.

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